Assignment 2: Multi-Map Spatial Analytics in Python

Title

From Raw Spatial Data to Insightful Maps: A Python-Based Geospatial Visualization Assignment

Objective

You are hired as a junior geospatial data analyst. Your task is to build a small **map analytics portfolio** in Python using multiple real-world datasets. You must solve different spatial problems using different map types, justify your design decisions, and reflect on the strengths and weaknesses of each visualization.

This assignment is not only about plotting maps. It is about answering the question:

When is a map necessary, which map type is appropriate, and what distortions or misleading effects must be considered?

Learning Outcomes

By the end of this assignment, you should be able to:

1. Load, clean, merge, and visualize geospatial datasets in Python.
2. Choose appropriate map types for different spatial questions.
3. Explain when a choropleth, proportional symbol map, contour map, flow map, or cartogram is appropriate.
4. Use and compare projections.
5. Avoid misleading geographic representations.
6. Build reproducible geospatial analysis workflows.

Required Python Stack

You should use Python and are encouraged to use these libraries:

- `pandas`
- `geopandas`
- `matplotlib`
- `plotly` or `folium` for interactive maps
- `contextily` for basemaps

- `shapely`
- `pyproj`
- `numpy`
- `scipy`
- `networkx` for flow/network work
- `cartopy` or `geoplot` if desired

Optional:

- `geemap`
- `osmnx`
- `rasterio`
- `scikit-learn`

Deliverables

You must submit:

1. **Python notebook(s)** or `.py` scripts
2. **A report** in PDF format, 8–12 pages
3. **Exported map figures** in PNG format
4. **One interactive map** in HTML

Assignment Structure

Part 0 – Dataset Collection and Documentation

Students must collect at least **4 different datasets**, each serving a different map problem.

Minimum required dataset categories

They must use datasets from at least these categories:

1. **Polygon/area dataset**
   Example: states, districts, counties, census tracts
2. **Point dataset**
   Example: city coordinates, hospitals, earthquakes, pollution stations
3. **Flow/movement dataset**
   Example: migration, trade, transportation, commuting, airline routes

4. **Continuous spatial field dataset**
   Example: temperature, rainfall, pollution, elevation, wind

Suggested dataset sources

- Natural Earth
- US Census TIGER/Line
- World Bank
- NOAA
- NASA EarthData
- OpenStreetMap
- WHO / CDC
- Kaggle
- Our World in Data
- local government open data portals

Deliverable

A table with:

- dataset name
- source
- file type
- geometry type
- spatial extent
- variables used
- why selected

Part 1 – Projection Challenge

Problem

Create **the same world or national dataset** using **at least three different projections** and analyze how interpretation changes.

Required projections

Choose at least 3 from:

- Mercator
- Equirectangular
- Azimuthal

- Albers Equal-Area
- Lambert Conformal Conic
- Winkel Tripel or Robinson if available

Tasks

1. Plot the same dataset in 3 projections.
2. Explain what is preserved and what is distorted.
3. Identify which projection is most appropriate for:
    - navigation
    - thematic comparison by area
    - general educational world map

Example datasets

- world population by country
- greenhouse gas emissions
- country area vs population

Expected insight

Students should show that projection is not a cosmetic choice; it changes perception of area, distance, and direction.

Output

- 3 maps
- 1 comparison table
- 1 short discussion

Part 2 – Choropleth Map and Its Pitfalls

Problem

Build a choropleth map for a regional variable and then critique its limitations.

Example questions

- Which regions have highest disease rates?
- Which counties have highest unemployment?
- Which districts have highest literacy?
- Which countries have highest child mortality?

Tasks

1. Create a choropleth map using normalized data.
2. Use an appropriate classification method:
    - quantiles
    - equal interval
    - natural breaks
3. Compare at least two classification strategies.
4. Explain when the choropleth is misleading.
5. Discuss why absolute counts should usually not be mapped directly on choropleths.

Required analysis

Students must explicitly answer:

- Why should ratio/rate data be preferred over raw counts?
- How does region size affect interpretation?
- What visual bias appears when large but sparsely populated regions dominate?

Output

- 2 choropleths
- classification comparison
- short critique

Part 3 – Proportional Symbol Map

Problem

Visualize point-based or aggregated magnitude data using proportional symbols.

Example questions

- Which cities have the largest populations?
- Where are earthquake magnitudes highest?
- Which hospitals have the highest patient loads?
- Which airports have the highest traffic?

Tasks

1. Plot symbol locations correctly.
2. Scale symbol size proportionally to magnitude.

3. Explain whether symbol area or symbol radius was used.
4. Reduce overlap using transparency, jitter, zoomed inset, or interactivity.
5. Compare the proportional symbol map with a choropleth for the same variable where possible.

Required conceptual point

Students must explain why proportional symbol maps are often better than choropleths for **point phenomena** or unevenly distributed data.

Output

- 1 static map
- 1 interactive map
- short comparison paragraph

Part 4 – Flow Map / Movement Analysis

Problem

Model and visualize movement between places.

Example problems

Choose one:

- migration between regions
- airline routes
- commuting flows
- trade volumes
- refugee movement
- shipping routes
- student movement between cities

Tasks

1. Create a flow map with origin-destination pairs.
2. Encode magnitude using line width.
3. Encode direction using arrows, color gradients, or annotations.
4. Filter clutter by showing top N flows or thresholding.
5. Identify a hub-and-spoke structure, corridor, or regional cluster.

Advanced requirement

Students must compute at least one of:

- top inflow node
- top outflow node
- net flow
- centrality using `networkx`

Output

- 1 flow map
- 1 network summary table
- 1 interpretation section

Part 5 – Continuous Field Mapping

Problem

Create a map for a continuous spatial variable using interpolation or contouring.

Example datasets

- rainfall stations
- temperature stations
- air pollution measurements
- elevation samples
- groundwater depth
- wind speed stations

Tasks

1. Start with point measurements.
2. Produce:
    - a point map
    - an interpolated surface or contour/isopleth map
3. Use interpolation such as:
    - IDW
    - grid interpolation
    - radial basis
    - scipy interpolation
4. Explain the assumptions and limitations.

5. Show where interpolation may be unreliable.

Required concept

Students must explain why contour/isopleth maps are suitable for **continuous fields** but not for arbitrary region-based categorical data.

Output

- point map
- contour/isopleth map
- interpretation

Part 6 – Cartogram or Alternative "Area Correction" Map

Problem

Demonstrate how standard geographic size can distort interpretation, then fix it using an alternative.

Example themes

- election results
- population by region
- emissions by country
- disease burden
- GDP by country
- parliamentary seats

Tasks

1. Show a standard geographic map.
2. Show an alternative representation:
    - cartogram
    - tile grid map
    - hex map
    - equal-area tile layout
3. Compare the message each map gives.
4. Discuss "size vs data effect."

Required concept

Students must show that land area does not equal importance, people, votes, or emissions.

Output

- original map
- alternative corrected map
- 1-page critique

Part 7 – Case-Based Map Design Challenge

Students must solve **three mini-scenarios** by selecting the correct map type and implementing it.

Scenario A – Public Health

A health department wants to know:

- where the disease burden is highest
- whether cases cluster
- whether the outbreak is spreading

Students must choose the best map type(s) and justify them.

Scenario B – Urban Services

A city wants to evaluate service accessibility to hospitals, schools, and bus stops.

Students must decide whether to use:

- point map
- service area map
- proportional symbols
- density map
- flow map

Scenario C – Climate Risk

A researcher wants to compare flood risk and exposed population across districts.

Students must combine at least two layers:

- hazard layer
- population layer
- optionally a choropleth or cartogram

Deliverable

For each scenario:

- chosen map type
- one reason why
- one reason why another map type would be worse
- implementation in Python

Minimum Technical Requirements

Students must demonstrate all of the following:

- read at least 4 different spatial datasets
- perform at least 2 dataset joins
- handle at least 2 geometry types
- reproject data at least 3 times
- create at least 6 distinct maps
- save high-resolution outputs
- include legends, titles, source labels, and coordinate/projection notes
- document assumptions and limitations

Suggested Dataset Combinations

Option 1 – U.S. Civic + Population + Weather

- county shapefile
- election/unemployment/population CSV
- city points
- NOAA weather station points
- migration or commuting flows

Option 2 – Global Population + Emissions + Flights + Climate

- world countries shapefile

- population by country
- emissions by country
- major airport coordinates
- airline routes
- temperature or rainfall data

Option 3 – Public Health + Mobility

- administrative boundaries
- disease rate by district
- hospital locations
- patient movement or ambulance trips
- air quality or weather stations

Grading Rubric (200 Points)

1. Data Acquisition and Cleaning – 25 points

- appropriate datasets chosen
- correct preprocessing
- joins and cleaning documented

2. Projection Understanding – 20 points

- correct reprojection
- thoughtful justification of projection choice

3. Choropleth Analysis – 20 points

- correct variable normalization
- good legend/classification
- critique of limitations

4. Proportional Symbol Map – 30 points

- correct scaling
- overlap handling
- clear interpretation

5. Flow Map – 20 points

- meaningful OD mapping
- readable design
- quantitative interpretation

6. Continuous Field / Contour Map – 20 points

- valid interpolation
- correct reasoning for continuous field mapping

7. Cartogram / Alternative Area Correction – 20 points

- good comparison with standard map
- clear critique of land-area bias

8. Scenario-Based Design Decisions – 25 points

- correct map type selection
- strong reasoning

9. Report Quality – 10 points

- clarity, structure, figure captions, citations

10. Code Quality and Reproducibility – 10 points

- readable code
- modular design
- reproducible workflow

Extra Credit Ideas

Students can earn bonus marks for any of the following:

- build an interactive dashboard
- add hover tooltips
- use animation for temporal flows
- compare multiple basemaps
- create a live web map

- implement a custom tile map or hex map
- use OSMnx for routing/network analysis
- compute spatial autocorrelation (Moran's I)

Recommended Submission Structure

Folder layout

```
assignment/
│
├── data/
├── notebooks/
├── scripts/
├── outputs/
│   ├── figures/
│   └── interactive/
├── report.pdf
└── README.md
```